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State-Based Process Monitoring in Traditional Event Logs for Concept Drift Detection

Bachelor Thesis

presented by

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Acknowledgments

If you want to incorporate it, keep in mind that, according to the official regulations, *independent work* is a very important aspect of writing a thesis. Thus, you can be grateful for the emotional support by your favorite pet, yet we recommend you to be careful if you want to write anything more specific. In general, we suggest to omit this section.

Abstract

The abstract should be a concise summary of what you have done. It should cover, in roughly 1 to 2 sentences per topic:

- The domain, i.e., explain the application domain in which your thesis is applicable.
- The problem, i.e., motivate what the problem is you are solving.
- The method, i.e., describe the method you have proposed, and, compare it, briefly & high level to other approaches. In particular, focus on the benefits of your approach versus the other approach(es)
- The evaluation, i.e., how did you evaluate your method, what results did you obtain and what do the results indicate?

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1 Introduction

First of all, we encourage you to read Prof. van der Aalst’s guide on how to write papers [9].

In this section, you introduce the work to your audience. Even though this is the first chapter of your thesis, you should not necessarily write this chapter first. It is often easier to write this chapter at the end of the thesis writing. In this part of your introduction, i.e., the part before the Motivation section (Section 1.1), you write a paragraph (not just a sentence) for each of the bullet points mentioned in the abstract:

- Domain, i.e., explain the application domain in which your thesis is applicable.
- Problem, i.e., motivate what the problem is you are solving.
- Method, i.e., describe the method you have proposed, and, compare it, briefly & high level to other approaches. In particular, focus on the benefits of your approach versus the other approach(es)
- Evaluation, i.e., how did you evaluate your method and what do the results indicate?

Optionally, you can finalize this part with a little overview of what is still to come in this chapter: “The remainder of this chapter is structured as follows. In Section 1.1, we motivate the need for ... In Section 1.2, we provide a formal problem statement. ...”

Before we dive into the motivation section, a few general guidelines for writing:

1. Write *active* and in the *present tense*.
2. Avoid “optionality” as much as possible, i.e., usage of the following words should (pun intended) be minimized:
 - could
 - would
 - should
 - may
 - might
 - can
 - will
 - ...

1 Introduction

Good: We present a method that ...

Bad: We will present a method that ...

Good: We use function f and compute its inverse.

Bad: We can use the function f and will compute its inverse.

3. Do not use `\\` to generate line breaks, but an empty line
4. Position the caption of a Figure below the figure.
5. Position the caption of a Table above the table.
6. Write Section Headings and Titles Like This
Good: Approximation Bias in Unstable Systems
Bad: Approximation bias in unstable systems
7. Be consistent in your references.
 - Make sure that the name of the same author is always the same (using DBLP helps for this)
 - Titles should be consistent. Either use the style previously described, i.e., Approximation Bias in Unstable Systems or Approximation bias in unstable systems (this is allowed in the references, opposed to your own section headings and titles, however, *BE CONSISTENT*).

1.1 Motivation

Played-out event logs do not change, but real processes do. Following van der Aalst [8], the situation in which a process is changing while being analyzed is called a concept drift. For example, two activities are concurrent at the start of an event log but then become sequential. According to van der Aalst [8], typical triggers of concept drift are seasonal effects, such as higher demand in December, and changed market conditions. Classical process mining techniques mostly assume that the process at the start and at the end of the log is the same [3]. If a drift occurs, process models or key performance indicators (KPIs) computed over the entire log mix two different processes. The process owner must therefore detect whether and when the process changed, and what changed.

A drift does not restrict itself to the order of activities. Processes run in a context, but process analysis sometimes ignores it. As described by van der Aalst [8], process context covers how cases influence each other like competing for the same resources. A single case is therefore not sufficient to explain the observed behavior alone. To explain the observed behavior and its changes, we analyze dependencies within cases, between cases, and with resources. For these dependencies, Senderovich et al. [7] introduce the terms intra-case and inter-case dependencies. This thesis adopts this terminology and considers a process from three perspectives:

- *Intra-case perspective*: The history of a single case up to a specific time, covering which activities happened in which order and how often. Drift example: A new trace variant dominates.

- *Resource perspective:* The working style of resources, such as their workload, service times, handovers, and the activities they perform. Drift example: A resource no longer exists and other resources take over the work.
- *Inter-case perspective:* The current situation of all running cases of the process, covering arrivals, waiting times, and case attributes. Drift example: The mean waiting time between consecutive events of a case increases.

From all three perspectives concept drifts can take place, but they can also influence each other. For example, a new trace variant with a frequent activity becomes dominant (intra-case perspective). The resource responsible for that activity then has more work to do (resource perspective), which affects the waiting times in the entire process (inter-case perspective).

We describe each drift by three properties:

- *Drift type:* What changes, e.g., the control-flow, the reassignment of an activity to another resource, or the waiting time between consecutive events.
- *Drift perspective:* The perspective to which the drift type belongs. For example, control-flow drifts belong to the intra-case perspective, reassignment drifts to the resource perspective, and waiting time drifts to the inter-case perspective.
- *Drift mode:* How the change takes effect over time, e.g., as a sudden, gradual, or recurring change.

In the literature, the terms drift type, drift mode, and drift perspective are used interchangeably. We reserve the term drift type for what changes, the term drift mode for how it changes, and the term drift perspective for the group a drift type belongs to.

Existing concept drift detection approaches cover this picture only partially:

- *Control-flow focus:* Most of the work on drift detection focuses on control-flow. Out of the 38 drift detection papers in the survey of Sato et al. [6], only six focus on the time perspective, three on the data perspective, and none on the resource perspective. In our framework, waiting times and case attributes fall under the inter-case perspective, and service times under the resource perspective. A concept drift that only changes the time or resource perspective stays undetected with control-flow approaches.
- *Single KPIs:* Approaches beyond control-flow often track single KPIs. Adams et al. [1] turn each KPI chosen by the user into a time series over time windows and detect drifts in each series separately. A drift therefore shows up as separate change points in several KPIs. This approach does not deliver a compact description of the overall process situation. For steady-state detection, Kraus et al. [4] show that combining complementary process characteristics is more accurate than relying on a single one.

1 Introduction

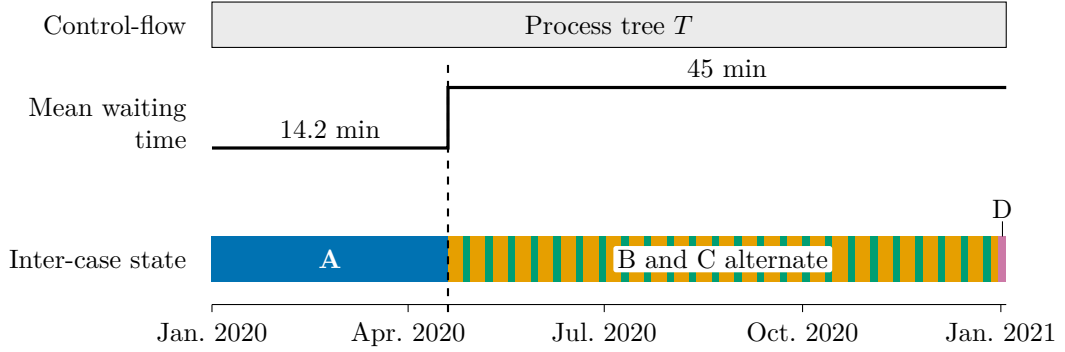


Figure 1.1: Visualization of the synthetic Rheon log generated with a mean waiting time drift. At the drift point (dashed), the mean waiting time between consecutive events of a case rises from 14.2 to 45 min, while all cases still follow process tree T . The inter-case state of each 12-hour window switches exactly there from A to an alternation of B and C. State D marks the run-out of the log after the last case arrivals. The alternation of B and C is drawn schematically.

- *Drift modes:* How a change takes effect over time is also important information for the process owner. Most of the existing approaches focus on sudden drifts only [6].
- *Missing test data:* Sato et al. [6] argue that real-world logs are unsuitable for measuring detection accuracy, since their drift points are unknown. They find only three public synthetic datasets with injected drifts. These datasets cover only the control-flow and time perspectives and do not mix different drift modes. None of them contains a resource drift or a change in case arrivals. A recent benchmark by Padella and Vinci [5] injects drifts into five real logs, but again only into the control-flow and activity durations.

Figure 1.1 shows this gap using a synthetic event log generated by Rheon. Rheon is our generator for synthetic event logs with injected concept drifts in all three perspectives (Chapter 4). This log has 300,299 cases with 1,946,540 events, seven activities, and seven resources. Its cases arrive between January 1 and December 31, 2020. On April 19, 2020, the mean waiting time between consecutive events of a case jumps from 14.2 minutes to 45 minutes. The activities, the control-flow, the resources, and the dominant resource of each activity remain unchanged. A detector that only compares relations between activities sees the same control-flow before and after the drift point and misses the change.

The state-based approach of this thesis aims to detect drifts in all three perspectives and to show what changed. Instead of separate change points per KPI, our states combine many features into one compact description. Rheon provides test logs with known drift points in all three perspectives. In Berti et al. [2], execution states from object-centric event logs are computed for objects. We adapt the approach from objects to cases in our case-centric framework and define our intra-case states. As an extension

to their work, our approach also takes into account resource and inter-case perspectives and adapts the idea to them.

Overall, we compute states from an event log by extracting features, compressing, and clustering. The input of our approach is an event log, and the output consists of intra-case, resource, and inter-case states whose changes over time indicate drifts. In Figure 1.1 one can visually see the general idea of our approach for the inter-case perspective. We describe each 12-hour window by features such as the number of active cases, arrivals, completions, and the mean waiting time between consecutive events of a case. Following the execution states of Berti et al., a principal component analysis (PCA) compresses these features. A 2×2 self-organizing map (SOM) is trained on these compressed feature vectors. Each of its four cells is one state, and each window gets the state of its best-matching cell.

The pipeline is unsupervised and does not see the drift point or the drift type. As seen in Figure 1.1, all 219 windows before the drift point fall into state A. From the drift point until December 31, the windows alternate between states B and C. The last seven windows, in which the log runs out, form state D. The first principal component explains 51% of the variance and is dominated by the mean waiting time between consecutive events of a case. It therefore also shows what changed: after the drift point, more time passes between the events of a case.

1.2 Problem Statement

An event log records a process over a long period, and the process often changes within this period. Such a concept drift affects the intra-case, resource, or inter-case perspective, or several of them at once. A detector that only captures control-flow drifts therefore misses drifts that only affect the resource or inter-case perspective.

Van der Aalst [8] names three main challenges in dealing with concept drifts:

1. *Change point detection*: Did the process change? If yes, when did it change?
2. *Change localization and characterization*: What has changed?
3. *Change process discovery*: How to capture and predict second-order dynamics?

According to van der Aalst, a process is not static even when there is no drift (first-order dynamics). This makes it even harder to detect changes of the process itself (second-order dynamics). Most approaches compare sliding windows of events with statistical methods. To localize drift points exactly and detect them at different time scales, the window length must vary. Moreover, drifts at different time scales can overlap, as in Figure 1.2.

This thesis focuses on challenges 1 and 2. To characterize a drift, our framework determines its drift perspective, its drift mode, and what exactly changed.

In conclusion, this thesis focuses on the following problem:

1 Introduction

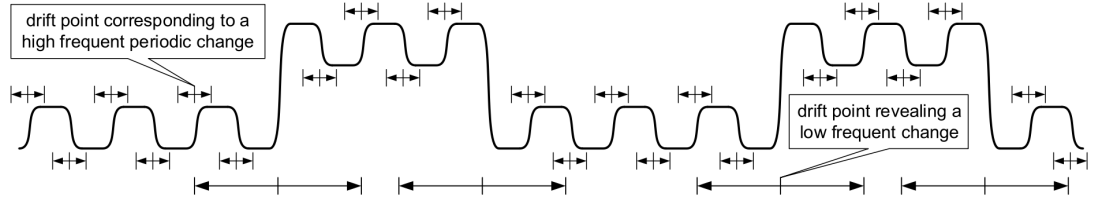


Figure 1.2: A periodically changing process with drifts at two different time scales, taken from van der Aalst [8]. Shorter sliding windows are used to detect short-term drifts. The left half of each sliding window is compared with the right half to spot statistically relevant changes. The larger sliding windows at the bottom are used for long-term drifts.

- *Given:* An event log in which each event belongs to exactly one case. Each event carries a case ID, an activity, and a timestamp, and optionally a resource and case attributes. Drift points, drift types, drift perspectives, and drift modes are unknown.
- *Objective:* Compute intra-case, resource, and inter-case states whose changes over time indicate drifts. From these changes, detect the drift points and determine for each drift its perspective, its mode, and what changed.
- *Evaluation:* We compare the detected drift points with known drift points in two sets of logs. First, synthetic Rheon logs contain drifts in all three perspectives. Second, the benchmark of Padella and Vinci [5] injects sudden, gradual, and recurring control-flow and duration drifts into five real logs.

1.3 Research Questions

The main research question of this thesis is:

Can a process be monitored as a combination of intra-case, resource and inter-case states, and is this combination a useful indicator for a concept drift in traditional case-centric event logs?

We divide our main research question into five further questions:

- RQ1** *States:* How can we compute execution states from a traditional event log in the intra-case, resource and inter-case perspectives, and how interpretable are these states in explaining concept drifts?
- RQ2** *Detection:* Do the changes over time in those states signal drift points even if an event log has drifts in multiple perspectives?

- RQ3** *Characterization*: Do the states indicate the perspective and the mode of a drift, and which features suit which drift types?
- RQ4** *Clustering*: Which of the clustering algorithms SOM, k-means and density-based spatial clustering of applications with noise (DBSCAN) delivers the most robust and reliable states?
- RQ5** *Applicability*: In which processes and event logs is the state-based drift detection approach more reliable and when does it deliver deceptive results?

RQ1 addresses the first part of the main research question, monitoring a process through states. RQ2 to RQ5 cover the second part of the main research question, the usage of the states as an indicator for concept drift detection.

1.4 Research Goals

In order to answer the research questions, we follow seven research goals. Each goal lists the research questions it addresses in parentheses.

- RG1** *Terminology*: To clearly define drift type, drift mode and drift perspective (RQ3).
- RG2** *Test data*: Develop a generator for synthetic event logs that injects concept drifts of various drift types in two drift modes (sudden and gradual). Along with each log, the generator writes a metadata file that lists the injected drifts (RQ2, RQ3, RQ5).
- RG3** *Framework*: Develop a state-based framework for concept drift detection (RQ1, RQ2). It adapts execution states from Berti et al. [2] for objects to cases and extends them to resource and inter-case states.
- RG4** *Library*: Develop an AI-extended Python library that implements our framework (RQ1, RQ4). The library offers each result as a plot or text for large language models (LLM) and also provides LLM connections.
- RG5** *Dashboard*: Develop a user-friendly dashboard around the library (RQ1, RQ3). In the dashboard, users compute and analyze results like the states and try different pipelines and configurations with the help of an LLM-based assistant, the Copilot.
- RG6** *Evaluation*: Evaluate the robustness and flexibility of the framework against different processes, event logs and timespans (RQ2, RQ5)
- RG7** *Limitations*: Reveal the limitations of the approach and potential misleading results especially when using the Copilot (RQ5)

1.5 Contributions

This thesis has six main contributions. Each contribution lists its research goals and chapters in parentheses.

- *Terminology*: A terminology defining and clearly separating drift types (what changes), drift modes (how it changes over time), and drift perspectives (which perspective the drift type belongs to) (RG1).
- *Rheon*: (RG2, Chapter 4). A library that generates synthetic event logs and injects nine drift types from all three perspectives, each suddenly or gradually. For each generated log, Rheon creates a metadata file comprehensively describing how the log was generated and a list of drifts together with their drift points, types and metadata.
- *State-based framework*: A framework that computes the intra-case, resource and inter-case states (RG3, Chapters 5 to 7). The framework consists of four steps: feature extraction, dimensionality reduction with PCA, clustering into states, and drift signals. The drift signals compare state distributions of calendar windows with divergences, or state vectors with similarity measures.
- *kairo*: A Python library that implements the framework (RG4, Section 8.1). With the help of its user-friendly documentation, users create a pipeline configuration in kairo and run it. The library supports various features addressing different drift types, various scaling techniques, PCA, and SOM, k-means and DBSCAN for state clustering. It also computes state frequency distributions for different window sizes, offers multiple divergences and similarity measures, and provides various plotting options for the different pipeline steps.
- *Dashboard*: A dashboard built around kairo (RG5, Section 8.2). In a user-friendly user interface (UI), users can upload an event log, run the pipeline steps for each perspective and investigate the states in detail. With the LLM-based Copilot, users ask questions, gain insights about the results and request further steps.
- *Evaluation*: An evaluation on synthetic Rheon logs with different drift combinations from all perspectives and on the five real event logs of Padella and Vinci [5] (RG6, RG7, Chapters 9 and 10). The evaluation encompasses comprehensive comparisons between SOM, k-means and DBSCAN, and between different configuration parameters like scaling techniques, window sizes and feature selections.

1.6 Thesis Structure

The remainder of the thesis is structured as follows. Chapter 2 reviews the literature on concept drift detection and state-based process mining approaches and places this thesis within the literature. Chapter 3 introduces the notation for event logs, foundations of

concept drifts, PCA and clustering. Chapter 4 presents Rheon, which generates the synthetic event logs for the evaluation.

Chapters 5 to 7 define intra-case, resource and inter-case states and answer RQ1. Each of these chapters follows the same steps: feature extraction, dimensionality reduction, clustering, and drift signals. Chapter 8 describes the implementation of kairo and the dashboard tool.

Chapter 9 answers RQ2 to RQ5 using the Rheon logs and the benchmark logs of Padella and Vinci [5]. Chapter 10 discusses the design decisions and limitations of the framework. Chapter 11 summarizes the results and discusses future directions and potential extensions of the topic.

2 Related Work

In this section, you are going to list all the related work. In this template, the related work section is located directly after the introduction chapter, however, this is not always the best/most logical location. As a general guideline, the position of the related work section can be:

- *Directly after the introduction*; Position the related work section here, in case you do not need to refer to any detailed (mathematical) concepts. Furthermore, you do not need to refer to any part of your solution. Positioning the related work section here is most commonly done if you are really uncovering “new terrain”, i.e., you are doing something *very novel*.
- *Directly after preliminaries*; In this scenario, you swap the related work and preliminaries, i.e., w.r.t. this thesis template. Position this here when you do need to refer to some of the (mathematical) concepts (typically presented in the preliminaries), yet, you do not need to use any properties of your own solution when comparing your work to the related literature.
- *Directly before the Conclusion (typically after the Evaluation)*; Position the related work section here, in case you need to compare the related work against properties of your algorithm and/or the results you have obtained.

In some cases, you can introduce sub-sections in this chapter, i.e., if a division of the related work eases the readability.

Note that the related work is not intended for you to “Convince your supervisors that you have studied a lot of papers”. Rather, you simply describe in 1, max 2 lines what the core idea of a paper is, how it relates to your work and why it is different from your work. In some cases, it is possible to discuss the aforementioned points for a selection of papers rather than a single paper. Try to present the paper along logical lines, i.e., define categories of work and introduce subsections for this. Never present a long 2-page list of randomly/chronologically structured work. Always add a decent structure!

3 Preliminaries

In the preliminaries you present well-understood mathematical concepts that you need in your thesis. For example, you can define the natural numbers as $\mathbb{N} := \{1, 2, \dots\}$, and, correspondingly $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. A preliminary notion is either a well-defined commonly understood mathematical notion, e.g., sets, multisets, graphs, sequences, Petri nets, $\dots$, or, it is a concept clearly defined in another paper, i.e., you just adopt the notation (or a slight variation thereof). *Any concept you use should be defined in your thesis.* You should never write: “We use Workflow nets, a definition of these can be found here [X]”. If you use it, explain it.

Concepts that are unique to your approach are not part of the preliminaries, i.e., they are described in the approach section itself.

Some useful tips:

- When introducing a complex concept, use the following structure (always works):
 - Explain the concept informally.
 - Provide a formal definition of the concept.
 - Provide an example, using the formal definition, of the concept.

In your examples, try to be *as visual as you can*, often, an image says more than 5 pages of text.

- use commands, some of which are already provided in the preamble, e.g.,
`\newcommand*{\N}{\mathbb{N}}`

4 Rheon

Placeholder.

Table 4.1: Placeholder table.

Column A	Column B
Placeholder	Placeholder

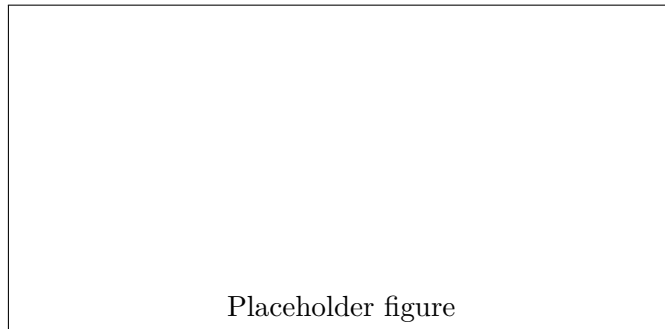


Figure 4.1: Placeholder figure.

5 Intra-Case State

Placeholder.

5.1 Feature Extraction

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5.2 Dimensionality Reduction

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5.3 State Clustering

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5.4 Drift Signals

Placeholder.

6 Resource State

Placeholder.

6.1 Feature Extraction

Placeholder.

6.2 Dimensionality Reduction

Placeholder.

6.3 State Clustering

Placeholder.

6.4 Drift Signals

Placeholder.

7 Inter-Case State

Placeholder.

7.1 Feature Extraction

Placeholder.

7.2 Dimensionality Reduction

Placeholder.

7.3 State Clustering

Placeholder.

7.4 Drift Signals

Placeholder.

8 Implementation

Placeholder.

8.1 Kairo

Placeholder.

8.2 Dashboard

Placeholder.

9 Evaluation

In this chapter, you present your evaluation. You should have extensively discussed with your supervisor what you are evaluating and why ...

Some typical sections that you need here:

9.1 Experimental Setup

Describe the setup of your experiments. If you apply a pipeline of different techniques, show the pipeline. Present the parameters of your experiments in a table, and, describe them. Typical Elements:

- Input Data Set(s) Used
- Algorithm(s) Used
- Parameter(s) Used
- ...

9.2 Results

Show the results. Try to present your results as structured as you can. A good result (sub)section, follows the following structure:

- Describe what results you are going to Show
- Present an initial hypothesis about the results, e.g., We expect the quality metric M to behave like this, conditional to this parameter P.
- Show the Results
- Confirm the hypothesis.
- If there are results that are not according to the hypothesis, you have to be able to explain why this is the case!!!

9.3 Threats to Validity

Here you discuss any element of your experiments (usually depending on the setup), e.g., data sets used, parameters assessed, that might have implications for the validity of your results. For example, if you have not considered a specific range of parameter values, it is unclear how the algorithm will behave for these settings. If you have excluded certain data, this might have its implications for the generalizability of your results. In a way, this is the discussion section of your thesis.

10 Discussion

In this section, you discuss your general approach. What design decisions did you make, and, what are the implications of this? Maybe your approach computes an exact intermediary result, which is costly in terms of time. Does a suitable approximation approach exist? What are the implications for that approach. You can also dive deeper in very related work and explain why it does not work in certain cases.

11 Conclusion

Your conclusion actually follows the same structure as your abstract. However, you present what the reader has seen, again, using the same concepts as you have already used in your abstract.

Future Work. Furthermore, you present interesting directions for future work!

A Some additional stuff

Try to avoid appendices. If you can't, this is a place to show all kinds of results that are supportive, yet, not critical for your thesis.

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